

No. of Functional Features Included in the Solution

Date	6 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)
1	ML Predict Engine	Scores applicant financial profile inputs dynamically using the active Decision Tree ML algorithm.	Backend scoring engine (/api/predict)	Done
2	Delinquency Check	Checks client past-due logs and enforces hard-coded override boundaries to reject delinquency records.	Backend Compliance Overrides	Done
3	Analyst Form Console	Provides parameter entry forms and renders explainable dial weights and risk factor lists.	Frontend Dashboard View	Done
4	Customer Pre-Check	Runs eligibility soft inquiries without triggering bureau credit scoring drops.	stepped Eligibility Wizard	Done

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)
5	Batch Auditor Scanner	Allows drag-and-drop uploads of credit applications in CSV formats to audit flags concurrently.	Compliance Portal Module	Done
6	Model Evaluator Matrix	Plots statistical indices (accuracy, F1, ROC-AUC) comparing all 4 ML algorithms in bar graphs.	Frontend Dashboard Charts	Done
7	Analyst Logs Ledger	Displays transaction histories of previous queries in session-persistent ledgers.	Data Storage Cache	Done
8	Offline Simulator	Implements mock rules algorithm in Javascript if the python Flask backend is offline.	Frontend Client Layer	Done

Feature Summary:

Metric	Count / Value
Total Features Planned	8 Features
Total Features Implemented	8 Features
Core / Must-Have Features	6 Features (ML Engine, Delinquency Check, Analyst Portal, Pre-qualification Check, Batch Audit, Log Ledger)
Additional / Nice-to-Have Features	2 Features (Stepped eligibility wizard tips, Local offline client simulation mode)
Features Tested & Verified	8 Features (100% verified via Locust load scripts and manual test specs)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Stepped Eligibility Wizard, Analyst Form Console, Navigation tabs, Status Gauges	HTML5 input forms, custom backdrop-blur glassmorphic CSS controls, dynamic SVG/Chart.js dials.
2	Backend / Logic	Machine Learning Inference server routing, classifier evaluation comparisons	Python 3.12 Flask REST controller, Gunicorn WSGI workers, model pre-loading routines.
3	Database / Storage	Serialized classifier binary caches, categorical encoders, and session storage history lists	model.pkl weights storage, preprocessor.pkl scaling metrics, browser-local storage ledger logs.
4	API / Integration	Dynamic JSON prediction endpoints and bulk CSV audit loaders	Flask POST routes (/api/predict, /api/predict_batch), Watson ML cloud API connector.
5	Security / Authentication	Underwriter role authorization barriers and automatic default overrides	Session token credentials, active Delinquency check overrides (auto-blocking status codes 1-5).

System Implementation and Deployment Notes:

The functional modules identified in this breakdown represent a production-ready, decoupled design. By encapsulating preprocessing and inference logic inside Scikit-learn pipelines, the backend Flask instance remains highly performant and can be updated independently of the frontend user views. The visual layouts are optimized for desktop resolutions using vanilla CSS Grid structures, while the in-memory pickle caching ensures fast response times under concurrent workloads. The delinquent compliance filters act as a fail-safe firewall that overrides predictive model estimates whenever a high-risk history profile is processed, ensuring compliance with banking risk management policies.